Insight into future mobility technology

❖ Introduction to Electric vehicles

An electric vehicle (EV) is a vehicle powered by an electric motor, instead of an internal combustion engine (ICE), and the motor is run using the power stored in the batteries. The batteries have to be charged frequently by plugging into any main (120 V or 240 V) supply. EV has a much longer history than most people realize. EVs were seen soon after Joseph Henry introduced the first DC-powered motor in 1830. The first known electric car was a small model built by Professor Stratingh in the Dutch town of Groningen in 1835. The first EV was built by in 1834 by Thomas Davenport in the U.S., followed by Moses Farmer, who built the first two-passenger EV in 1847. There were no rechargeable electric cells (batteries) at that time. An EV did not become a viable option until the Frenchmen Gaston Plante and Camille Faure respectively invented (1865) and improved (1881) the storage battery.

EVs are known as zero emissions vehicles (ZEVs) and are much environment friendly than gasoline- or LPG-powered vehicles. As EVs have fewer moving parts, maintenance is also minimal. With no engine there are no oil changes, tune-ups, or timing and there is no exhaust. EVs are also far more energy efficient than gasoline engines and they are very quiet in operation.

EVs have been in continual use since the 1900s in various applications. Today these quiet vehicles with no tailpipe emissions are no longer limited to golf carts. New advances in battery technology, system integration, aerodynamics, and research and development by major vehicle manufacturers has led to the producing of electric vehicles that will play a practical role on city streets.

❖ Engineering philosophy of EVs

Engineering integrates science, technology and management to solve real world problems effectively and economically, while usually asks for ‘how’ and ‘what for’, pursues ‘best solution’ and emphasizes ‘teamwork’. As EV engineering is multidisciplinary, specialists in those areas of engineering must work together; electrical engineers, electronic engineers, mechanical engineers, chemical engineers and mainstream automotive engineers must pool their knowledge in the main areas that must be integrated: body design, electric propulsion, energy sources, energy refuelling and energy management.

❖ EV CONCEPT

The modern EV concept is summarized as follows:

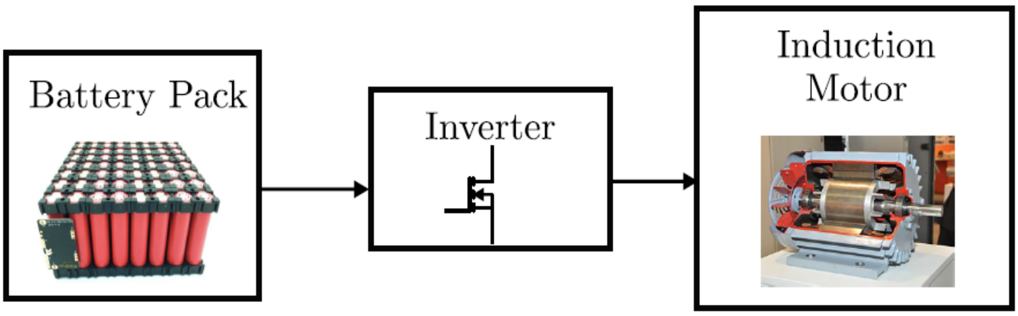
- The EV is a road vehicle based on modern electric propulsion which consists of the electric motor, power converter and energy source, and it has its own distinct characteristics;
- The EV is not just a car but a new system for our society, realizing clean and efficient road transportation;
- The EV system is an intelligent system which can readily be integrated with modern transportation networks;
- EV design involves the integration of art and engineering;
- EV operating conditions and duty cycles must be newly defined;
- EV users' expectations must be studied, hence appropriate education must be conducted.

❖ What are Electric Vehicles (EVs)?

Electric vehicles use an electric motor for propulsion rather than using an internal combustion engine. Electric vehicles have a battery that can be charged by electric supply. This electric energy is used to run the motor. Electric vehicles can be fully electric or hybrid type, which means a combination of electric motor and combustion engine.

An Electric Vehicle contains 3 main parts

(1) Energy Source. (2) Power Converter. (3) Traction Motor.



Battery Pack-Energy source

The most common battery type in modern electric vehicles are lithium-ion and lithium polymer, because of their high energy density compared to their weight. Other types of rechargeable batteries used in electric vehicles include lead–acid ("flooded", deep-cycle, and valve regulated lead acid), nickel-cadmium, nickel–metal hydride, and, less commonly, zinc–air, and sodium nickel chloride ("zebra") batteries.[1] The amount of electricity (i.e. electric charge) stored in batteries is measured in ampere hours or in coulombs, with the total energy often measured in kilowatt-hours.

Inverter

An inverter is a device that converts DC power to the AC power used in an electric vehicle motor. The inverter can change the speed at which the motor rotates by adjusting the frequency of the alternating current. It can also increase or decrease the power or torque of the motor by adjusting the amplitude of the signal.

Motor

Using power from the traction battery pack, this motor drives the vehicle's wheels. Some vehicles use motor generators that perform both the drive and regeneration functions.

Both AC and DC motors use electrical current to produce rotating magnetic fields that, in turn, generate rotational mechanical force in the armature—located on the rotor or stator—around the shaft. The various motor designs use this same basic concept to convert electric energy into powerful bursts of force and provide dynamic levels of speed or power.

Main Motor Components

Electric and hybrid vehicles are more complex than traditional internal combustion engine (ICE) vehicles in some ways, due to the integration of electric components. Here are some key components found in electric and hybrid vehicles:

Electric Vehicles (EVs)

Electric Motor: Replaces the internal combustion engine for propulsion.

Battery Pack: Stores energy for the electric motor to use. Usually lithium-ion batteries are used.

Inverter: Converts DC (Direct Current) from the battery pack to AC (Alternating Current) for the electric motor.

Electric Vehicle Supply Equipment (EVSE): Charging infrastructure, including wall chargers, fast chargers, and other charging stations.

Power Electronics Controller: Manages the flow of electrical energy between the battery, inverter, and electric motor.

Regenerative Braking System: Converts kinetic energy back into stored energy in the battery during braking.

Thermal Management System: Controls the temperature of the battery and other electronic components.

Vehicle Control Unit (VCU): The "brain" of the vehicle that manages various subsystems.

High-voltage Switches and Relays: Safety and control systems for the high-voltage battery pack.

Auxiliary Systems: These include traditional systems found in ICE vehicles like air conditioning and power steering, but they are electrically powered in an EV.

Telemetry and Connectivity Modules: Systems for connecting the vehicle to networks for updates, diagnostics, and sometimes remote control.

❖ Need of Electric Vehicles (EVs) Or Importance of Electric vehicle

• Carbon monoxide, sulphur dioxide, hydrocarbons are some of the gases which cause air pollution. These gases are the result of emission from an internal combustion engine. According to a report by WHO, India is home for 14 out of 20 most polluted cities in the world.

• All these pollution-causing gases and emission is harmful to people like women, children and elderly people. This emission will also lead to global warming.

• On the other side electric vehicles are environmentally friendly and do not emit toxic gases, it will help reduce the global warming effect.

• It will help to fulfil the commitment of the Paris climate agreement.

• It will be helpful in reducing oil imports. India is 5th largest importer of oil with consumption of 2.2 million barrels a day.

• Electric vehicles are cost-effective in maintenance (because of less fluid and less moving parts). Government is also providing subsidies on vehicles and lower taxes on an electric motor.

❖ Some drawbacks of Electric Vehicles

• Electric engines provide limited mileage. A complete recharge will take a lot of time and fewer infrastructures for charging across the country.

• It will be poor for highway driving; the top speed of the electric car is 70mph.

• Due to the high cost of lithium-ion battery and the use of electricity will make the overall cost of the electric vehicle high.

• A small amount of pollution is also caused by electric vehicles because of the toxicity of batteries and all electricity origin is also not renewable.

❖ Challenges related to the EV industry in India

• Electric vehicles are not penetrated in the Indian market, almost the lowest in the world.

• India is having challenges related to technological production of electronics such as the battery, semiconductors etc.

• Government's uncertainty in policymaking is another challenge for investors in this industry. It is a capital-intensive sector and will make a profit in the near future.

• The local production is just 35% of the total input production of electric vehicles.

• India does not have any reserve for lithium and cobalt which is important for battery production which is the most important component for EVs. This is related to increasing the cost of production because of dependency on China and Japan.

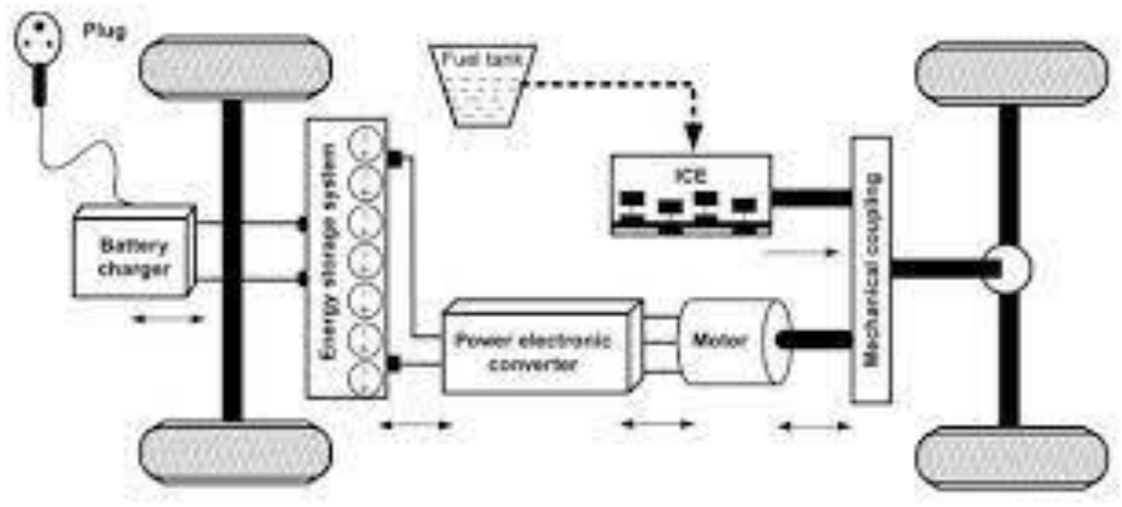
• High rate of GST and depreciation of rupee is another challenge for the EV industry.

• Lack of infrastructure related to AC versus DC charging station, grid stability, fear anxiety-related to the battery will run out soon, are other hindrances.

• This industry needs a skilled worker and high services.

• India lacks dedicated courses and training in this area.

Hybrid Vehicles



Hybrid vehicles contain components from both internal combustion engine vehicles and electric vehicles.

Internal Combustion Engine (ICE): Standard gasoline or diesel engine for propulsion and generating electricity.

Electric Motor: Provides additional propulsion and works in tandem with the ICE.

Battery Pack: Usually smaller than in a fully electric vehicle, it provides energy storage for electric-only driving and assists the ICE.

Power Split Device: Manages power distribution between the ICE and the electric motor.

Transmission: Could be Continuously Variable Transmission (CVT) or automated manual, depending on the design.

Inverter and Converter: Manage electrical energy conversion and distribution.

Regenerative Braking System: Similar to the system in an EV, it stores energy during braking.

Thermal Management System: Manages temperature for both the ICE and electrical systems.

Control Unit(s): Manages the interplay between the electric and ICE components for optimized efficiency.

Fuel Tank: Stores fuel for the ICE, just like in traditional vehicles.

Auxiliary Systems: Similar to those in both EVs and ICE vehicles, like air conditioning, but can be powered by either the ICE, electric motor, or both.

Telemetry and Connectivity Modules: Similar to those in EVs.

These components are integrated in various ways depending on the type of hybrid vehicle, which could be a mild hybrid, full hybrid, or plug-in hybrid. Each type has different capabilities and limitations in terms of electric-only driving range, fuel efficiency, and other parameters.

Regenerative Braking System For Automobile

Every time we step on our car's brakes, we are actually wasting energy. As we know that energy can neither be created nor be destroyed. It can be just converted from one form to another. So when our car slows down, the kinetic energy that was propelling it in the forward direction has to go somewhere. Most of it simply gets released in the form of heat and becomes useless. That energy, which could have been used to do work, is essentially wasted. The solution for this kind of this problem is Regenerative Braking System.

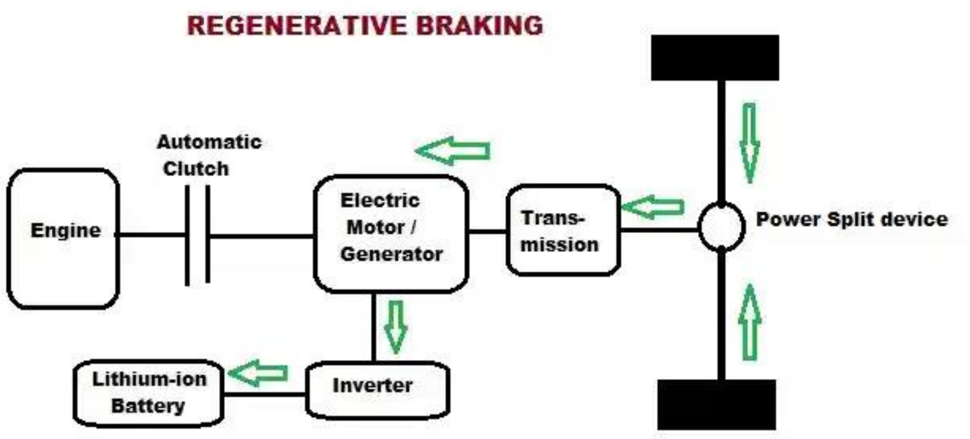
This is a new type of braking system that can recollect much of the car's kinetic energy and convert it into electrical energy or mechanical energy. Regenerative braking is one of the emerging technologies of automotive industry which can prove to be very beneficent. Using regenerative braking system in a vehicle not only results in the recovery of the energy but it also increases the efficiency of vehicle (in case of hybrid vehicles) and saves energy, which is stored in the auxiliary battery.

WORKING PRINCIPLE:

We have to fabricate this model, when you are applied the brake condition while some energy is going loss that is conditions to avoiding, we have to fabricate on kinetic energy storage method, which is contain for like sun and planetary gear and scroll spring also attached.

When you need the brake, your applied the brake and both of clutch also applied. Brake condition done stop the wheel and disc contain system and clutch condition is disengaged the wheel and axle that is condition energy going loss, whereas our model engage the ring gear is rotate the reverse direction, as well as connecting on scroll spring rotate also same direction that is condition save the kinetic energy on clutch applied condition.

When you release the clutch, regenerating the kinetic energy on axle which is condition increased the pickup and fuel efficiency.



ADVANTAGE

- Improved Performance.
- Improved Fuel Economy - Dependent on duty cycles, power train design, control strategy, and the efficiency of individual components.
- Reduction in Engine wears.
- Reduction in Brake Wear- Reducing cost of replacement brake linings, cost of labor to install them, and vehicle down time,
- Emissions reduction- engine emissions reduced by engine decoupling, reducing total engine revolutions and total time of engine operation.

- Operating range is comparable with conventional vehicles- a problem not yet overcome by electric vehicles.

APPLICATION:

- For recovering Kinetic energy of vehicle lost during braking process.
- One theoretical application of regenerative braking would be in a manufacturing plant that moves material from one workstation to another on a conveyer system that stops at each point.
- Regenerative braking is used in some elevator and crane hoist motors.
- Regenerative Braking Systems are also used in electric railway vehicle (London Underground & Virgin Trains).